INCH-POUND

MIL-T-10163E(ME)
8 January 1992
SUPERSEDING
MIL-T-10163D
24 August 1981

MILITARY SPECIFICATION

# TERRAIN MODELING KIT, TRAINING AID

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a terrain model training aid consisting of base and hilltop terrain, miniatures, maps, mapping symbols, paint dispenser, rollers, colors, brushes, crayons, sponges, and a chest with tray to hold the above items.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-558 - Maptacks.
H-B-371 - Brush, Artist's, Marking.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6910

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| | |
|---|---|
| L-S-626 | - Sponges, Synthetic. |
| UU-P-258 | - Paper, Index. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-C-843 | - Cushioning Material, Cellulosic. |

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| MIL-W-530 | - Webbing, Textile, Cotton, General Purpose, Natural or in Colors. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-C-12044 | - Chest, Plywood. |
| MIL-P-43027 | - Paper, Chart, Map and Book, Lithographic Finish, Coated and Uncoated. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1190 | - Minimum Guidelines for Level C Preservation, Packing and Marking. |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government Drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

| | |
|---|---|
| TA13220E9950 | - Terrain Modeling Kit, Training Aid. |
| 13220E9959 | - Terrain Model Training Aid Miniature Mapping Symbols. |
| 13220E9960 | - Terrain Model Training Aid, Paint Dispenser Roller, Roads and Railroad Tracks. |

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, and Engineering Center, ATTN: STRBE-JDT, Fort Belvoir, VA 22060-5606.)



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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The terrain model training aid shall be furnished as a complete kit. Construction details and quantities of the individual items in the kit shall be as shown on TA13220E9950, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the terrain model training aid. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the



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applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The terrain model training aid shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the terrain model training aid may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the terrain model training aid may be newly fabricated from recovered materials to the maximum extent practicable, provided the terrain model training aid produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the terrain model training aid.

3.4 Webbing. Webbing shall conform to MIL-W-530, type II, class 4.

3.5 Paint. Paint for plastic, rubber, or urethane foam shall be rubber base and shall be compatible with items on which it is used.

3.6 Base and hilltop sections. The base and hilltop sections shall be urethane foam plastic or rubber. The sections shall be formed to portray the contours of the mountains, hills, valleys, and general terrain conditions, as applicable. The urethane foam shall be self-skinning and the construction of the sections shall be one stage. Rubber and foam rubber construction of the sections shall consist of a shell or skin of rubber approximately 0.125 inch thick. The shells shall be held in undistorted relationship by a backing or reinforcement of foam rubber permanently secured to the underside of the shell. A tolerance of $\pm$0.0625 inch is permissible at matching edges of the base sections, provided such mismatching is not caused by curl or wave in the sides of the sections or by cave or distortion of the surface of the sections. Each section shall be imprinted with a part number as listed on the drawings.



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3.6.1 Color scheme and application of paint. The exposed surfaces of the section shells shall be painted in colors as specified herein.

3.6.1.1 Airbrush spray painting sequence. Airbrush spray painting shall be performed in the following sequence.

a. The entire surface of the model, including sides, shall be sprayed opaque yellow.
b. Green shall be sprayed very lightly over the yellow, with the valleys and sides of hills being sprayed more heavily to produce a contrast between flat areas and molded areas; sides of sections shall be lightly sprayed.
c. Brown shall be sprayed over the green using the same very light spray over the entire surface, with the valleys and sides of hills being sprayed darker.
d. In spraying the green over the yellow and the brown over the green, the under color shall show through; that is, these colors shall not be applied opaque. Since all sides of the four base sections are to be matched to each other, care should be taken to produce a consistently even color above all matching edges.

3.6.1.2 Brush painting. All water areas shall be painted with a brush.

3.6.1.3 Color composition. The color composition shall be as specified herein. Evidence of these compositions shall be certified by the manufacturer.

a. The composition of yellow ground tone (airbrush) shall be as follows:

| Composition | Percent |
|---|---|
| White | 83 |
| Raw Sienna | 15 |
| Ultramarine blue | 1 |
| Burnt umber | 1 |

b. The composition of green (airbrush) shall be as follows:

| Composition | Percent |
|---|---|
| Light chrome yellow | 67 |
| Ultramarine blue | 15 |
| Burnt umber | 11 |
| Burnt sienna | 7 |



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c. The composition of brown (airbrush) shall be as follows:

| Composition | Percent |
|---|---|
| Raw sienna | 64 |
| Burnt sienna | 18 |
| Burnt umber | 9 |
| White | 9 |

d. The composition of blue (hand brush) shall be as follows:

| Composition | Percent |
|---|---|
| Turquoise | 95 |
| Burnt sienna | 5 |

3.6.1.4 Tolerance on colors. A slight variation in coloring from one set of four base sections to another set is permissible; however, within each set, coloring shall be uniform and carefully controlled.

3.7 Landscape miniatures. Trees, tree clumps, hedgerows, and shrubbery clumps shall be of cellular flexible material having a rubber covering approximately 0.0625 of an inch thick with a integral dark green color representing foliage. Painting the items after application of the rubber will not be acceptable. Individual trees shall have a corrosion-resistant metal pin inserted in the base of the trunk.

3.8 Miscellaneous miniatures. Equipment, personnel, structures, and roadblocks shall be made of a smooth-finishing firm tough material that is capable of producing sharp details. Each miniature shall be painted as indicated on the applicable drawing. The water tower shall have corrosion-resistant metal legs with sharpened ends and the individual infantry personnel shall have a corrosion-resistant metal pin inserted in the legs.

3.9 Maps. Maps shall be printed on paper conforming to MIL-P-43027, type I. Twelve sets of maps shall be furnished. Each set shall consist of 8 sheets, 30-inch by 30-inch trim size, which shall be full scale copies of the base and hilltop contour drawings. Hilltop sections shall be printed two sections per sheet only. Each sheet shall include contour lines with title "Terrain" or "Hill" as applicable, and part numbers.

3.10 Symbol cards. A set of 25 symbol cards (3 Artillery, 4 Armor, and 18 Infantry) as depicted on Drawing 13220E9959 of the Top Assembly, shall be furnished. Symbol cards shall be printed on 220-pound white index card stock conforming to UU-P-258, grade B.

3.11 Maptacks. A box of 100 maptacks, conforming to A-A-558, type I, 0.3125 inch metal head, 1.25 inches long, shall be furnished.



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3.12 Sponges. Two sponges, conforming to L-S-626, type II, porosity B, class 1, size 3, shall be furnished.

3.13 Brushes. Two marking brushes conforming to H-B-371, Size 15/64 inch, shall be furnished.

3.14 Paint dispensing unit. The paint dispensing unit, fabricated and supplied in accordance with 13220E9960 of the Top Assembly, shall operate properly, and the rollers shall produce clean prints on both paper and rubber base sections.

3.15 Tempera colors. Tempera colors shall be furnished in 2-ounce screwtop glass jars and shall be compatible with the items on which they are used. The following colors shall be furnished:

| Color | Quantity |
|---|---|
| Ultramarine blue | 3 jars |
| Raw umber | 3 jars |
| Dark deep green | 2 jars |
| Yellow | 2 jars |

3.16 Crayons. Crayons shall be artist's pastel, 1/2-inch diameter, 3 inches long. Four crayons each of the following colors shall be furnished: ultramarine blue, raw umber, dark green, and yellow.

3.17 Chest. The chest shall be in accordance with drawing 13220E9940 of the top assembly, and with MIL-C-12044. The color of the paint shall be forest green, lusterless in accordance with MIL-T-704. The chest shall ensure a stable loading arrangement of the components and, when properly set up, shall provide a level and sturdy working table in the field.

3.18 Identification marking. The terrain model training aid shall be identified in accordance with MIL-STD-130.

3.19 Workmanship. All components of the terrain model training aid shall be constructed to insure proper functioning and be free of defects which would affect their appearance and impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform



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any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article terrain model training aid shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article terrain model training aid shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.



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4.4.3 Tests. Sampling selected as specified in 4.4.1 shall be tested as specified in 4.5.2, except that not less than two terrain model training aids shall be selected. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The terrain training aid shall be examined as specified herein for the following defects:

101. Components missing or not as specified (see 3.1).
102. Materials not as specified (see 3.3).
103. Material deterioration prevention and control not as specified (see 3.3.1).
104. Dissimilar metals not protected against galvanic corrosion as specified (see 3.3.2).
105. Identification of materials and finishes not as specified (see 3.3.3).
106. Recovered materials not as specified (see 3.3.4).
107. Webbing does not conform as specified (see 3.4).
108. Components not painted as specified (see 3.5).
109. Base and hilltop sections not as specified (see 3.6 thru 3.6.1.4).
110. Metal pins and legs not corrosion resistant (see 3.7 and 3.8).
111. Maps not as specified (see 3.9).
112. Symbol cards not as specified (see 3.10).
113. Maptacks not as specified (see 3.11).
114. Sponges not as specified (see 3.12).
115. Brushes not as specified (see 3.13).
116. Tempera colors not as specified (see 3.15).
117. Crayons not as specified (see 3.16).
118. Chest not as specified (see 3.17).
119. Identification marking missing, incomplete or illegible (see 3.18).
120. Workmanship not as specified (see 3.19).

4.5.2 Tests.

4.5.2.1 Assembly test. The chest with the individual assembled items shall be closed and placed on its ends and sides and then reopened and examined. Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.2 Setup test. The chest shall be set up in table position by disengaging the hinge and unfolding the legs and shall be examined. Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.3 Operational test. The paint dispenser unit shall be assembled and filled with any one of the four tempera colors. The dispenser shall then be tested on both paper and rubber base sections using three different rollers. Nonconformance to 3.14 shall constitute failure of this test.



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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

| | |
|---|---|
| 114. | Preservation not as specified for level A or level C (see 5.1). |
| 115. | Components not placed in the chest and the chest not placed in the box as specified for level A (see 5.1.1). |
| 116. | Closure and sealing of box not as specified for level A (see 5.1.1). |
| 117. | Packing not as specified for level A, B or C (see 5.2). |
| 118. | Strapping not as specified for level A (see 5.2.1). |
| 119. | Marking missing, illegible, incorrect, or incomplete (see 5.3). |

5. PACKAGING

5.1 Preservation. Preservation shall be level A or level C, as specified (see 6.2).

5.1.1 Level A. The components of the terrain model training aid shall be placed in the chest in accordance with the loading plan. Cushioning material conforming to PPP-C-843, type I, class A, shall be utilized as required, to prevent movement or damage to the contents in the chest. The chest shall then be placed in a close-fitting box conforming to PPP-B-636, W5c, style optional. Folded pads of fiberboard shall be placed within the box to prevent the hinges, hasp, or handles from puncturing the box. Closure and sealing of the box shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Level C. The components of the terrain model training aid shall be placed in the chest in accordance with the loading plan and preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, level B, or level C, as specified (see 6.2).

5.2.1 Level A. Each chest shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, or conforming



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to PPP-B-621, class 2, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification except that strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A, except the box shall be domestic type and strapping shall not be required.

5.2.3 Level C. The components of the terrain model training aid shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking for shipment and storage shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The terrain model training aid is intended for use in troop training and in the study of military tactics.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval (see 3.2).
d. Level of preservation and level of packing required (see 5.1 and 5.2).
e. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior



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Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Subject (key word) listing.

Urethane foam (self skinning)
Color scheme
Painting sequence
Color composition
Loading arrangement

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing Activity:
Army - ME

Project 6910-A038



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-T-10163E(ME) | 2. DOCUMENT DATE (YYMMDD)<br>92/01/08 |
|---|---|---|

3. DOCUMENT TITLE

Terrain Modeling Kit, Training Aid

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| | (703) 664-5717 | 354-5717 |
| c. ADDRESS (Include Zip Code)<br>US Army Belvoir RDE Center<br>ATTN: STRBE-TSE | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 | |

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